Ladhe's Quad Conjecture: Every Prime Quadruplet Is an Additive/Multiplicative/Exponential Combination of Its Predecessors*

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Abstract

Prime quadruplets are the constellations $(p, p+2, p+6, p+8)$ of four primes, the densest possible packing of four primes beyond $(5, 7, 11, 13)$. We order them into a chain $\mathcal{Q}_1 = (11, 13, 17, 19)$, $\mathcal{Q}_2 = (101, 103, 107, 109)$, $\mathcal{Q}_3 = (191, 193, 197, 199)$, ..., seeded by the *royal quad* $\mathcal{Q}_0 = (2, 3, 5, 7)$, and observe that every member of every quadruplet can be written as the largest member of the previous quadruplet plus a sum of distinct members of earlier quadruplets, for example $13001 = 9439 + 3461 + 101$ and $35545421 = 35524459 + 18049 + 2089 + 821 + 3$. The precise form of the observation is that the gap D between a quadruplet member and its base is a sum of at most 8 distinct earlier members, with exactly ten exceptional gaps, all at most 382, that need a small expression in the royal members $2, 3, 5, 7$; allowing one such expression in every gap brings the bound down to six terms with no exception at all. We call this *Ladhe's Quad Conjecture* and verify it for all 28,387 prime quadruplets below 10^9, i.e. for 113,544 quadruplet members, with an independent checker. Two further representations of each gap, one that prefers multiplication and one that prefers powers of chain members, are also computed and released. We test the objection that the pattern is a consequence of density rather than of primality by running the same solver on control chains of identical spacing whose members are not prime: the bound of eight and the parity of the term counts are reproduced by a control that preserves the residue class, and are therefore structural, but no control reproduces the small closed set of exceptions. We also prove, from a two-line lemma and computed base cases, that the chain is an additive basis for all integers: every integer up to the next quadruplet (indeed up to a bound that grows like the sum of all earlier members) is a sum of distinct chain members plus one expression in $2, 3, 5, 7$ with $+, -, \times$; this is unconditional below 10^9 and rests beyond that only on a mild growth condition. We explain carefully what the conjecture does and does not say about locating the next quadruplet, give a heuristic for why such representations are plentiful, and state open questions. The utility that builds the chain and the full dataset are available at `https://github.com/SPAlgorithm/LE`.

1 Introduction

A *prime quadruplet* is a quadruple of primes of the form

$$(p,\ p+2,\ p+6,\ p+8),$$

*Correspondence: `spalgorithm@gmail.com`. Paper and dataset are archived on Zenodo, `https://doi.org/10.5281/zenodo.22286846` (this DOI always resolves to the latest version), and the tightest admissible pattern of four primes. Apart from $(5, 7, 11, 13)$, every prime quadruplet has $p \equiv 11 \pmod{30}$, so its members end in the digits $1, 3, 7, 9$ in that order. The sequence of first members $11, 101, 191, 821, 1481, 1871, 2081, \ldots$ is OEIS A007530 [3]. Whether there are infinitely many prime quadruplets is open; it is the case $k = 4$ of the Hardy–Littlewood prime k-tuple conjecture [1], and the Bateman–Horn heuristic [2] predicts that the number of quadruplets up to x is asymptotically $C_4\, x/(\log x)^4$ with $C_4 \approx 4.15$. The strongest unconditional results in this direction are the bounded-gap theorems of Zhang [4] and Maynard [5], which give infinitely many pairs (and m-tuples) of primes within a bounded distance but not the specific pattern $(0, 2, 6, 8)$. Large quadruplets and related clusters have been searched for computationally for decades [6].

This note records an additive structure that runs through the whole sequence of prime quadruplets, at least as far as we have looked. Order the quadruplets by size and call the k-th one $\mathcal{Q}_k$; put the four primes $2, 3, 5, 7$ in front as $\mathcal{Q}_0$, the *royal quad*. Then every member of $\mathcal{Q}_k$ is obtained from the largest member of $\mathcal{Q}_{k-1}$ by adding a few distinct members of earlier quadruplets. Section 3 states this precisely in three forms of increasing strength, Section 4 proves a companion statement, that with subtraction allowed among the royal members every integer below the next quadruplet is such a sum, Section 5 walks through examples that can be checked by hand, Section 6 reports the computation up to 10^9, and Section 8 discusses, with some care, what the statement means for finding the next quadruplet.

The observation grew out of the authors' earlier work on additive decompositions of primes, *Ladhe's Conjecture* [7], which concerns single primes written as sums of three primes and underlies a hash-based signature scheme [8]. The present note is about prime *quadruplets* and is self-contained.

2 Definitions and rules

Definition 1 (Quads). *$\mathcal{Q}_0 = (2, 3, 5, 7)$ is the* royal quad*. For $k \geq 1$, $\mathcal{Q}_k$ is the k-th prime quadruplet $(p, p+2, p+6, p+8)$ with $p \geq 11$, in increasing order of p. We write its members as $q_k^{(1)} = p$, $q_k^{(3)} = p+2$, $q_k^{(7)} = p+6$, $q_k^{(9)} = p+8$, the superscript being the last decimal digit. The quadruplet $(5, 7, 11, 13)$ is not part of the chain, because 5 and 7 already belong to $\mathcal{Q}_0$, and $\mathcal{Q}_0$ itself is not of the form $(p, p+2, p+6, p+8)$.*

Definition 2 (Base, gap, pool). *For $k \geq 2$ the* base *of $\mathcal{Q}_k$ is $m_k = q_{k-1}^{(9)}$, the largest member of the previous quadruplet. For a member t of $\mathcal{Q}_k$ the* gap *is $D = t - m_k > 0$. The* pool *available to $\mathcal{Q}_k$ is $\mathcal{P}_k = \mathcal{Q}_1 \cup \cdots \cup \mathcal{Q}_{k-1}$, the set of all $4(k-1)$ primes of the earlier quadruplets.*

Definition 3 (Royal expressions). *A* royal expression *is an arithmetic expression built from the members $2, 3, 5, 7$ of $\mathcal{Q}_0$, each used at most once, with the operations $+$ and $\times$ only. Let $\mathcal{R}$ be the set of its values; $\mathcal{R}$ has 68 elements between 2 and 210. Two subsets matter below: $\mathcal{R}_+$, the 11 values reachable by addition alone, $\{2, 3, 5, 7, 8, 9, 10, 12, 14, 15, 17\}$, and $\mathcal{R}_{\text{flat}}$, the 38 values with a* flat *expression, one in which no product contains a sum, such as $(5 \times 7) + (2 \times 3) = 41$ but not $5 \times (7 + (2 \times 3)) = 65$.*

Definition 4 (Additive representation). *An* additive representation *of a member t of $\mathcal{Q}_k$ ($k \geq 2$) is an identity*

$$t \;=\; m_k \;+\; s_1 + s_2 + \cdots + s_j \;+\; r, \qquad s_1 > s_2 > \cdots > s_j, \;\; s_i \in \mathcal{P}_k, \;\; r \in \mathcal{R} \cup \{0\},$$

with all s_i distinct and none equal to m_k. Its length *is j if $r = 0$ and $j+1$ otherwise. Several s_i may come from the same quadruplet. (The stricter reading "at most one member per quadruplet" already fails at $\mathcal{Q}_2$: $101 - 19 = 82$, and no single member of $\mathcal{Q}_1$ leaves a remainder in $\mathcal{R}$.) A member t of $\mathcal{Q}_1$ is written as a royal expression alone, e.g. $11 = (2 \times 3) + 5$.*

The following congruences are elementary but shape everything below. All first members satisfy $p \equiv 11 \pmod{30}$, so consecutive first members differ by a multiple of 30, and the gaps of the four members of $\mathcal{Q}_k$ satisfy

$$D \equiv 22,\ 24,\ 28,\ 0 \pmod{30} \quad \text{for } t = q_k^{(1)}, q_k^{(3)}, q_k^{(7)}, q_k^{(9)} \text{ respectively.}$$

Every gap is even while every pool member is odd, so a representation with $r = 0$ has an even number of terms. Pool members are $\equiv 11, 13, 17, 19 \pmod{30}$, hence a two-term representation of the gap of a first member ($D \equiv 22$) must add two first members ($11 + 11 \equiv 22$).

Three ways. The utility described in Section 6 computes, for every member, one representation of each of three kinds:

- **A** *additive*, as in the definition above, chosen so that addition is exhausted before any royal multiplication is used (fewer multiplications first, then fewer terms, then larger members first);
- **M** *multiplicative*: any two distinct members of $\mathcal{P}_k \cup \mathcal{Q}_0$ may be multiplied, and multiplication is preferred over addition (the largest unused member is multiplied by the largest member that still fits, then the next, plain addition only when nothing can be multiplied);
- **E** *exponential*: in addition, a member may be raised to a member power, powers being preferred over products over plain addition. Because the order in which powers are tried changes the outcome, two variants are kept: E1 takes the largest base first, E2 the largest power value first;
- **M1** *without the base*: the M search applied to the member itself instead of to its gap. No base term is required, the pool is every member of $\mathcal{P}_k \cup \mathcal{Q}_0$ below the target, and the largest products carry the number, e.g. $854921 = (427249 \cdot 2) + (109 \cdot 3) + (17 \cdot 5) + 11$;
- **E3** *powers first*: the largest power b^e below the member, with b and e distinct members, then again the largest power below what is left as long as it covers at least half of it, followed by the remainder in the additive way (members below the target added, largest first, closed by a flat royal expression), e.g. $854921 = 829^2 + 11^5 + 17^3 + 1489 + 107 + 101 + 19$ and $19497221 = 11^7 + 2^{13} + 829 + 827 + 199 + 3$. If no choice of members closes the remainder, the chain of powers is shortened by one, and finally the next largest first power is tried.

In A, M, E1 and E2 every member of $\mathcal{P}_k \cup \mathcal{Q}_0$ is used at most once and the base m_k is always a plain term; M1 and E3 also use every member at most once, the base and exponent of the E3 power included, but have no base term.

3 The conjecture

We state the observation in three forms. The first is the one the utility works with; the second and third are the ones we consider interesting, because they bound the number of terms.

Conjecture 1 (Ladhe's Quad Conjecture, existence form)**.** *Every member of every prime quadruplet $\mathcal{Q}_k$ with $k \geq 2$ has an additive representation over the chain: $t = m_k + \sum_i s_i + r$ with distinct $s_i \in \mathcal{P}_k$ and $r \in \mathcal{R} \cup \{0\}$.*

Conjecture 2 (bounded pure form)**.** *There is an absolute constant K such that, apart from finitely many gaps, every gap $D = t - m_k$ ($k \geq 2$) is a sum of at most K distinct members of $\mathcal{P}_k$. The data support $K = 8$, with exactly ten exceptional gaps:*

$$D \in \{22,\ 82,\ 84,\ 88,\ 90,\ 172,\ 174,\ 178,\ 180,\ 382\},$$

each of which needs a royal value. No new *exceptional gap occurs beyond $\mathcal{Q}_{435}$ (first member* 3,512,231*) up to 10^9: the ten values are complete there and recur without addition, the last occurrence being at $\mathcal{Q}_{28063}$ (first member* 985,124,051*). In all,* 254 *of the* 113,544 *members meet an exceptional gap, spread over* 98 *quadruplets.*

Conjecture 3 (bounded form with one flat royal value)**.** *Every gap $D = t - m_k$ ($k \geq 2$) is a sum of at most five distinct members of $\mathcal{P}_k$ plus at most one element of $\mathcal{R}_{\text{flat}}$; that is, every member has an additive representation of length at most six whose royal part, if any, is flat.*

Remark 1 (Why the existence form alone is weak)**.** *Once a few quadruplets exist, the pool is large and the sums of its subsets cover every even number in range many times over, so Conjecture 1 is very likely to hold for reasons that have little to do with quadruplets; it would also be vacuous if only finitely many quadruplets existed. What is not automatic is that a* fixed *number of terms suffices while the gaps grow: the average gap between consecutive quadruplets near x grows like $(\log x)^4/C_4$, about 35,000 at 10^9, and the pool members that fit under a gap are few (92 below the average gap and 336 below the largest gap, 395,520, in our data). Conjectures 2 and 3 would be falsified by a single gap that is not a short sum; the natural falsification test is a very large gap between consecutive quadruplets.*

Remark 2 (The exceptional gaps)**.** *All ten exceptions are small and appear early. The four gaps $82, 84, 88, 90$ belong to $\mathcal{Q}_2$ (and reappear whenever two consecutive quadruplets are 90 apart, as at $\mathcal{Q}_3$, $\mathcal{Q}_{14}$, $\mathcal{Q}_{437}$, ...); at that point the pool is $\mathcal{Q}_1$ alone, whose subset sums cannot reach 82. The gaps 22, 172–180 and 382 are coverage holes of $\mathcal{Q}_1 \cup \mathcal{Q}_2$: for instance $382 = 1871 - 1489$ needs all of $109, 107, 103, 19, 17, 13, 11$ plus 3. With the 11 royal addition values allowed, only $82, 84, 88, 90$ remain exceptional; with $\mathcal{R}_{\text{flat}}$ allowed there is no exception. Appendix B lists the ten gaps with their forced representations.*

4 Coverage: every integer below the next quadruplet

The chain also serves as an additive basis for *all* integers, not only for the quadruplets, once subtraction is allowed inside the royal part. This section proves that statement from two computed base cases.

Definition 5 (Signed royal values)**.** *$\mathcal{R}^{\pm}$ is the set of non-negative values of expressions built from the members $2, 3, 5, 7$ of $\mathcal{Q}_0$, each used at most once, with $+$, $-$ and $\times$ placed anywhere. It has 83 elements between 0 and 210, contains every integer from 0 to 38, and its first gaps are $39, 43, 53, 59, 61, 62$. Examples: $0 = 5 - (3 + 2)$, $1 = 3 - 2$, $4 = 7 - 3$, $33 = (5 \times 7) - 2$, $55 = 5 \times ((2 \times 7) - 3)$.*

Definition 6 (Derivation)**.** *A* derivation *of an integer $n \geq 0$ over $\mathcal{Q}_0, \dots, \mathcal{Q}_k$ is an identity $n = s_1 + \cdots + s_j + r$ with distinct $s_i \in \mathcal{Q}_1 \cup \cdots \cup \mathcal{Q}_k$ and $r \in \mathcal{R}^{\pm}$ ($j = 0$ or $r = 0$ allowed). Quadruplet members are only added; subtraction and multiplication occur only among royal members. Write C_k for the set of integers that have a derivation over $\mathcal{Q}_0, \dots, \mathcal{Q}_k$ and T_k for the largest T with $[0, T] \subseteq C_k$.*

By direct computation,

$$T_0 = 38, \qquad T_1 = 156, \qquad T_2 = 576, \qquad T_3 = 1356, \qquad T_4 = 4656,$$

so with the pool $2, 3, 5, 7, 11, 13, 17, 19$ every integer from 0 to 156 has a derivation, in particular every integer up to 101; for instance $16 = 11+5$, $53 = 19+11+(7\times5)-2$ and $55 = 19+11+(7\times5)$.

Lemma 1. *Let A be a set of non-negative integers containing every integer in $[0, T]$, and let $0 < m \leq T + 1$. Then $A \cup (A + m)$ contains every integer in $[0, T + m]$.*

Proof. An integer $x \in [0, T]$ lies in A. An integer $x \in [T + 1, T + m]$ satisfies $0 \leq x - m \leq T$, so $x - m \in A$ and $x \in A + m$. □

Proposition 1. *Put $S_3 = T_3 = 1356$ and $S_k = S_{k-1} + 4p_k + 16$ for $k \geq 4$. If $p_j \leq S_{j-1} + 1$ for every j with $4 \leq j \leq k+1$, then $T_k \geq S_k \geq p_{k+1}$: every integer n with $0 \leq n \leq p_{k+1}$ has a derivation over $\mathcal{Q}_0, \ldots, \mathcal{Q}_k$. For $k \leq 3$ the same conclusion holds by the computed values ($T_0 \geq p_1 = 11$, $T_1 \geq p_2 = 101$, $T_2 \geq p_3 = 191$, $T_3 \geq p_4 = 821$).*

Proof. Induction on $k \geq 3$ with the claim $T_k \geq S_k$; the case $k = 3$ is the computation. Assume $T_{k-1} \geq S_{k-1}$ and add the members p_k, $p_k + 2$, $p_k + 6$, $p_k + 8$ of $\mathcal{Q}_k$ one at a time, each with coefficient 0 or 1. The first satisfies $p_k \leq S_{k-1} + 1 \leq T_{k-1} + 1$ by hypothesis, so Lemma 1 raises the coverage to $T_{k-1} + p_k$; each following member is at most $p_k + 8$, far below the coverage reached so far, so the lemma applies three more times and $T_k \geq T_{k-1} + 4p_k + 16 \geq S_k$. Finally $p_{k+1} \leq S_k + 1$ and p_{k+1} is odd while S_k is even, hence $p_{k+1} \leq S_k$. □

The hypothesis is mild: it fails only if a quadruplet's first member exceeds $S_{j-1} + 1$, which grows like four times the sum of all earlier first members. In the chain up to 10^9 it holds at every $j \leq 28{,}387$, and it is tightest at $j = 4$ ($p_4 = 821$ against $S_3 + 1 = 1357$). The one-step condition $p_j \leq 5p_{j-1} + 16$ is sufficient and also holds throughout (the largest ratio p_j/p_{j-1} for $j \geq 3$ is 4.30, at $j = 4$). So Proposition 1 is unconditional below 10^9; an unconditional proof for all k would follow from any Bertrand-type bound for prime quadruplets, which is open, although far more than what is needed here.

Remark 3. *Coverage is a statement about* all *integers with subtraction allowed and does not imply Conjecture 1, which forbids subtraction, fixes the base m_k and, in its bounded forms, limits the number of terms. It does show that the differences D of the conjecture, which are tiny compared with T_{k-1}, always have* some *derivation of this looser kind.*

The utility prints a derivation of any integer with `-d`, taking the largest quad prime below the remainder first, each prime once, and closing with a flat royal expression (no product containing a sum) whenever the choice of primes allows it, ranked by fewest multiplications, then fewest minus signs; a range `-d 53 83` prints one line per integer:

```
python3 lc.py -d 16        16 = 13 + 3
python3 lc.py -d 53        53 = 19 + 17 + 13 + 7 - 3
python3 lc.py -d 55        55 = 19 + 17 + 13 + 5 + 3 - 2
python3 lc.py -d 76        76 = 19 + 17 + 13 + 11 + (3*7) - 5
python3 lc.py -d 99        99 = 19 + 13 + (2*5*7) - 3
python3 lc.py -d 101       101 = 19 + 17 + 13 + 11 + (5*7) + (2*3)
python3 lc.py -d 999986171
                           999986171 = 999900529 + 82729 + 2089 + 823 + 3 - 2
```

The greedy choice is not the only derivation; $53 = 19+11+(7\times5)-2$ and $55 = 19+11+(7\times5)$ are equally valid. Up to 20,000 every integer receives a flat closing.

5 Worked examples

The lines below are copied verbatim from the utility (`python3 lc.py 12 -AME`); `A`, `M`, `M1`, `E1`, `E2`, `E3` label the ways, and a line is omitted when it repeats an earlier line of the same member (`E2` when it coincides with `E1`; in $\mathcal{Q}_1$, which has no base, `M1` coincides with `M`). Products are written in parentheses and $b \,\hat{}\, e$ means b^e.

$\mathcal{Q}_1 = (11, 13, 17, 19)$ **from the royal quad.** The first quadruplet is written with royal expressions only; the utility also lists alternatives, including ones with subtraction, which the chain equations of Section 3 otherwise never use.

```
11 = (2*3) + 5        also: (2*7) - 3  |  (3*7) - (2*5)  |  7 + 5 + 2 - 3
13 = (2*5) + 3
17 = 7 + 5 + 3 + 2
19 = (2*7) + 5
```

$\mathcal{Q}_2 = (101, 103, 107, 109)$**: the gap** 82**.** Base $m_2 = 19$, gap $101 - 19 = 82$. The pool is $\mathcal{Q}_1$, whose members other than the base sum to 41, so a royal value is unavoidable; 41 itself is flat, $(5 \times 7) + (2 \times 3)$.

```
A:  101 = 19 + 17 + 13 + 11 + (5*7) + (2*3)
M:  101 = 19 + (17*3) + (13*2) + 5
M1: 101 = (19*5) + (3*2)
E1: 101 = 19 + (7^2) + (11*3)
E3: 101 = (7^2) + 19 + 17 + 13 + 3
```

$\mathcal{Q}_3 = (191, 193, 197, 199)$**: the same gap again.** $191 - 109 = 82$, so the representation of the gap is reused with the new base. The utility stores one equation per distinct gap and reuses it. The base-free lines M1 and E3 are computed per member and cannot be reused; here they need only $\mathcal{Q}_1$ and the royal quad.

```
A:  191 = 109 + 17 + 13 + 11 + (5*7) + (2*3)
M:  191 = 109 + (17*3) + (13*2) + 5
M1: 191 = (19*7) + (17*3) + 5 + 2
E1: 191 = 109 + (7^2) + (11*3)
E3: 191 = (13^2) + (5*3) + 7
```

$\mathcal{Q}_4 = (821, 823, 827, 829)$**: pure addition.** Gap $821-199 = 622 = 197+193+191+19+17+5$; the three remaining members of $\mathcal{Q}_3$ are used together with two members of $\mathcal{Q}_1$ and the royal 5. The two exponential variants differ here: E1 starts from the largest base 19^2, E2 also, but then prefers $3^5 = 243$ over $5^3 = 125$.

```
A:  821 = 199 + 197 + 193 + 191 + 19 + 17 + 5
M:  821 = 199 + (197*3) + (13*2) + 5
M1: 821 = (199*3) + (103*2) + 13 + 5
E1: 821 = 199 + (19^2) + (5^3) + 101 + 17 + 11 + 7
E2: 821 = 199 + (19^2) + (3^5) + 11 + 7
E3: 821 = (19^2) + (7^3) + 101 + 11 + 5
```

$\mathcal{Q}_6 = (1871, \ldots)$**: a long gap early in the chain.** Gap $1871 - 1489 = 382$ is one of the ten exceptions: no subset of $\mathcal{Q}_1 \cup \mathcal{Q}_2$ sums to 382, so a royal value is needed. The shortest representation with a royal *addition* value has seven terms $(199+109+19+17+13+11+14$, with $14 = 7+5+2)$; with one flat royal value four terms suffice, $382 = 191+101+17+73$ with $73 = (2 \times 5 \times 7) + 3$. The utility's addition-first rule prefers the eight-term form below, whose royal part is the single member 3. The multiplicative way is one product.

```
A:  1871 = 1489 + 109 + 107 + 103 + 19 + 17 + 13 + 11 + 3
M:  1871 = 1489 + (191*2)
M1: 1871 = (829*2) + (19*7) + (17*3) + 13 + 11 + 5
E1: 1871 = 1489 + (19^2) + (7*3)
E3: 1871 = (11^3) + 199 + 197 + 109 + 17 + 13 + 5
```

$\mathcal{Q}_{12} = (13001, \ldots)$**: the typical two-term gap.** Gap $13001 - 9439 = 3562 = 3461 + 101$, one member of $\mathcal{Q}_9$ and one of $\mathcal{Q}_2$, both first members as the congruence $22 \equiv 11+11 \pmod{30}$ requires. About one gap in four has such a two-term representation.

```
A:  13001 = 9439 + 3461 + 101
M:  13001 = 9439 + (1489*2) + (193*3) + 5
M1: 13001 = (5659*2) + (199*7) + (19*13) + (5*3) + 17 + 11
E1: 13001 = 9439 + (19^2) + (13^3) + (109*7) + 107 + 101 + 17 + 11 + 5
E2: 13001 = 9439 + (13^3) + (19^2) + (109*7) + 107 + 101 + 17 + 11 + 5
E3: 13001 = (109^2) + 829 + 193 + 19 + 17 + 13 + 11 + (7*5) + 3
```

Two consecutive quadruplets near 3.5×10^7**.** $\mathcal{Q}_{2208}$ and $\mathcal{Q}_{2209}$ are 6,000 apart; the gap of the first, $35545421 - 35524459 = 20962$, is $18049 + 2089 + 821 + 3$, and the gap of the second, $35551421 - 35545429 = 5992$, is $5659 + 199 + 109 + 17 + 5 + 3$.

```
A:  35545421 = 35524459 + 18049 + 2089 + 821 + 3
M:  35545421 = 35524459 + (9439*2) + (199*7) + (193*3) + (19*5) + 17
M1: 35545421 = (17771269*2) + (829*3) + (19*13) + (7*5) + 103 + 11
E1: 35545421 = 35524459 + (109^2) + (19^3) + (199*7) + (107*5) + 193 + 101
E3: 35545421 = (5659^2) + (19^5) + (101^3) + 13009 + 1489 + 109 + 103 + 17 + 13

A:  35551421 = 35545429 + 5659 + 199 + 109 + 17 + 5 + 3
M:  35551421 = 35545429 + (2089*2) + (199*7) + (103*3) + (19*5) + 17
M1: 35551421 = (17773369*2) + (1489*3) + (19*11) + 7
E1: 35551421 = 35545429 + (19^2) + (17^3) + (103*5) + (13*7) + 101 + 11
E3: 35551421 = (5659^2) + (19^5) + (101^3) + 19423 + 823 + 193 + 191 + 103 + 7
```

The last quadruplet below 10^9 is $\mathcal{Q}_{28387} = (999986171, \dots)$ with $999986171 = 999900529 + 82729 + 2089 + 821 + 3$, and the largest gap between consecutive quadruplets in that range, 395,520 at $\mathcal{Q}_{19558}$, is still a two-term sum: $628641701 = 628246189 + 392261 + 3251$.

6 Computational evidence

The utility. `lc.py` is a single Python 3 file with no dependencies. It finds quadruplets with a segmented sieve (only $p \equiv 11 \pmod{30}$ is tested), then searches a representation of every member's gap by iterative deepening over the number of terms, with the preferences listed in Section 2, and stores one set of A/M/E1/E2 equations per distinct gap, plus the base-free M1 and E3 equations of every member, in a JSON cache that later runs reuse. The additive search is capped at 16 terms and 300,000 nodes per call, the M/E searches at 50,000 nodes, the base-free M1 search, whose pool is the whole chain, at 1,000,000 nodes, and the additive remainder of E3 at 300,000 nodes; a fallback path exists for the case that a search gives up, and it was never taken in the data reported here (zero fallback flags). Building the chain to 10^9 took 165 s on a 2019 Intel laptop under Python 3.13.5, of which the A/M/E1/E2 equations take 85 s; the base-free M1 and E3 equations, computed once per member over the whole chain, account for the rest.

The checker. `check.py` re-reads the JSON and, independently of the search code, verifies every stored equation: the arithmetic, that all members are distinct, that the base is never reused as a term, and that every term is smaller than the target. On the 10^9 dataset it checked 681,288 equations (6 per member for all 113,548 members, $\mathcal{Q}_1$ included) and found 0 errors.

Minimal lengths. `stats.py` recomputes, for every distinct gap and independently of the preferences of the utility, the minimal length K of a representation under three vocabularies: pool members only (*pure*); pool members plus at most one value of $\mathcal{R}_+$; pool members plus at most one value of $\mathcal{R}_{\text{flat}}$. The minimum is taken when the gap is first met, using only the quadruplets available at that moment. Table 1 gives the result.

Table 1: Minimal length K of a representation of the gap, over the 13,636 distinct gaps met below 10^9 (all four members).

vocabulary	K=1	2	3	4	5	6	7	8	none	mean
pure members	–	2817	–	10375	–	410	–	24	10	3.65
+ one value of $\mathcal{R}_+$	–	2818	429	9946	365	53	20	1	4	3.59
+ one value of $\mathcal{R}_{\text{flat}}$	5	2927	3333	7295	68	8	–	–	0	3.33

The pure column has only even lengths, as it must. The ten gaps without a pure representation are those of Conjecture 2; with $\mathcal{R}_+$ only $82, 84, 88, 90$ remain; with $\mathcal{R}_{\text{flat}}$ nothing remains and six terms always suffice, which is Conjecture 3. The mean minimal length does not grow along the chain: in eight consecutive blocks of about 3,550 quadruplets the mean pure K of the first member's gap is $3.80, 3.75, 3.73, 3.71, 3.71, 3.67, 3.73, 3.71$ and the share of two-term gaps stays between 22% and 24%.

What the utility actually stores. Table 2 summarises the additive representations chosen by the utility. Because addition is exhausted before any royal product is used, these are often longer than the minimum: only 120 of the 113,544 member equations contain a royal product (all with gap $82, 84, 88$ or 90), 52,790 end with an added royal value, and 60,634 are pure sums of pool members. The longest stored equation has nine terms after the base. Of the 28,386 first-member gaps, 24,977 repeat a gap seen earlier in the chain; there are only 13,636 distinct gaps among the 113,544 member gaps, because consecutive quadruplets are often the same distance apart.

Table 2: Additive representations stored by the utility, 10^9 dataset. Length counts the terms after the base, a royal value counting as one.

	length								royal part
scope	2	3	4	5	6	7	8	9	none / + / ×
first member of each quad	6493	–	19472	30	2173	119	99	–	14776 / 13580 / 30
all four members	26875	60	77222	60	8323	227	755	22	60634 / 52790 / 120

Multiplicative and exponential ways. For the 28,386 first-member gaps, the M representations use 90,661 products and 50,439 plain terms, never more than nine terms; 2,935 of the products multiply two royal members. E1 uses 60,176 powers, 49,728 products and 79,246 plain terms (at most 14 terms); E2 uses 63,213 powers, 34,739 products and 78,861 plain terms (at most 12). E1 and E2 coincide for 9,117 of the 28,386 gaps. These representations are released for further study; we make no claim about them beyond their existence within the search caps. The base-free M1 and powers-first E3 representations were likewise found for every one of the 113,548 members (only 11, 17 and 19, which are written from the royal quad alone, admit no power chain and fall back to the power-first search) and are stored and verified. The E3 chain has one power for 16,926 members, two for 35,775, three for 46,553 and four for 14,291; we do not analyse these representations further here.

7 Is this a property of primes, or of density?

A natural objection to Conjecture 2 is that the chain grows dense enough for representability to be forced by counting alone. By the time we reach $\mathcal{Q}_k$ the pool $\mathcal{P}_k$ holds $4(k-1)$ members, so the number of sums of at most eight distinct members is $\binom{4(k-1)}{8}$, vastly exceeding the range of D. On that reading the conjecture would say nothing about primes. This section separates what the objection explains from what it does not.

What is structural

Two features of the data are not evidence about primes, and we do not claim them as such.

The first is the parity of the term counts. Every representation observed uses an *even* number of pool members — the counts are $2, 4, 6, 8$ and never odd — and this is forced. Every member of every $\mathcal{Q}_j$ is an odd prime, so $D = t - m_k$ is a difference of two odd numbers and hence even; a sum of j odd numbers is congruent to j modulo 2; therefore j is even. No arithmetic beyond parity is involved.

The second is the bound $K = 8$ itself. To test it we ran the same solver on control chains built to have the same spacing as the real one, differing only in that their members are not prime:

- **jitter** — each quadruplet displaced inside its own local gap, preserving order and local density;
- **uniform** — each first member redrawn uniformly in the interval between its neighbours;
- **shift** — each first member displaced by a multiple of 30, preserving the residue class 11 mod 30, and with it the parity argument above, while destroying primality.

Every control keeps the shape $(q, q+2, q+6, q+8)$, the same counting function and the same base rule $m_k = q^{(9)}_{k-1}$, so any difference is attributable to the arithmetic of the members and not to how many there are. The **shift** control attains the same bound of eight terms and the same all-even distribution. The bound is a consequence of the residue class, not of primality.

What is not

chain	worst gap	exceptional gaps	occurrences	largest
real prime quadruplets	8	**10**	254	**382**
control, jitter (seed 1)	12	665	2334	1700
control, jitter (seed 2)	11	194	535	1193
control, uniform (seed 1)	10	613	17790	2416
control, uniform (seed 2)	11	152	12728	1334
control, shift	8	101	3057	3390

Table 3: All 28,387 quadruplets of each chain below 10^9, 113,544 members, under the base rule of Section 2. "Exceptional gaps" counts distinct values of D admitting no purely additive representation.

The exceptions are a different matter. The real chain has ten, the best control has 101 — an order of magnitude more — and every control admits exceptions several times larger than the real chain's maximum of 382. The contrast survives every control family we tried, including the one that reproduces the bound and the parity exactly.

Density and residue structure together predict that exceptions should be *rare*. They do not predict that there should be only ten, that all ten should be small, or that the set should close: no new exceptional gap appears after $\mathcal{Q}_{435}$, while 28,000 further quadruplets add none. That is the part of Conjecture 2 we regard as genuinely arithmetic, and the part we would most like to see explained.

8 What the conjecture says about the next quadruplet

It is tempting to read Conjecture 2 as a recipe: take the largest member of the last known quadruplet, add a short sum of earlier members, and the next quadruplet appears. We want to be precise about what is and is not true here.

What it gives. Every quadruplet after the first carries a *certificate* over its predecessors: a short equation whose terms are earlier chain members, with the base term identifying the previous quadruplet. The chain is therefore self-referential in a strong sense: nothing outside $\mathcal{Q}_0, \ldots, \mathcal{Q}_{k-1}$ is needed to write down $\mathcal{Q}_k$ once it is known. The certificate also turns the search for the next quadruplet into a bounded additive search: if the conjecture holds with K terms, then the first member of $\mathcal{Q}_k$ is of the form $m_k + D$ with $D \equiv 22 \pmod{30}$ and D a sum of at most K distinct pool members, and the candidate can be tested by four primality tests. The utility does exactly this in reverse, producing the certificate after the sieve has found the quadruplet.

What it does not give. The candidates produced by the additive search are far more numerous than quadruplets: near 10^9 the number of two-term sums of pool members below a typical gap is already in the thousands, and almost every admissible D has some short representation. So the conjecture does not point at the next quadruplet, it only says where it will be found once it exists, and in practice a sieve remains the faster way to find it. Nor does the conjecture imply that a next quadruplet exists; infinitude of prime quadruplets is a separate, open problem.

Why representations are plentiful. The Bateman–Horn heuristic predicts roughly $C_4\, y/(\log y)^4$ quadruplets below y, and so four times as many pool members. The number of j-element subsets of pool members below D grows like

$$\frac{1}{j!}\left(\frac{4C_4\, D}{(\log D)^4}\right)^j,$$

while their sums range over $O(D)$ even numbers, so for $j \geq 2$ the expected number of j-term representations of a given gap grows without bound as D grows, and grows faster the larger j is. This is the reason the mean minimal length stays flat, and it suggests the answer to the following question is yes.

Question 1. *Is every sufficiently large gap a sum of* two *distinct earlier members? Equivalently, do the lengths* $4, 6, 8$ *in Table 1 eventually disappear?*

Question 2. *Is a royal value ever needed again beyond the ten exceptional gaps, i.e. is Conjecture 2 true with exactly these exceptions and* $K = 8$*, or even* $K = 6$ *beyond some point?*

Question 3. *The same construction applies to any prime constellation, for instance twin primes or sexy prime triplets, with the corresponding admissible residues. Which constellations admit a bounded chain, and with what* K*?*

Question 4. *The exceptional gaps are exactly the coverage holes of* $\mathcal{Q}_1 \cup \mathcal{Q}_2$*. Is there a simple description of the set of gaps that are not sums of pool members at the moment they are first met, for the general constellation?*

9 Availability and reproducibility

The utility, the checker and the statistics script are released in the folder `LC` of the repository

https://github.com/SPAlgorithm/LE

The commands below rebuild everything reported here from scratch.

```
python3 lc.py --upto 999999999 --cache paper/qarray_1e9.json   # 85 s
python3 paper/check.py paper/qarray_1e9.json                   # 0 bad
python3 paper/stats.py paper/qarray_1e9.json                   # all figures
python3 lc.py 25 -AME --cache paper/qarray_1e9.json            # the examples
python3 paper/density.py 28387                       # Section 7, ~20 min
```

The dataset `qarray_1e9.json` (48,068,688 bytes) has SHA-256

`72dabebf238cfb792ccb4fc51d126128b497967d9e9b29d68077e7e0cd360477`

and `quads_1e9.csv` lists, for each of the 113,548 members (including $\mathcal{Q}_1$), the base, the gap, the four A/M/E1/E2 equations (the base-free M1/E3 equations are in the JSON only) and the three minimal lengths. The control chains of Section 7 are generated inside `density.py` from a fixed seed, so its table is reproducible without any further data. Interactive use: `python3 lc.py` asks for a number of quadruplets; `-M`, `-E`, `-AME` select the ways; `--all` shows all four members; `-v` shows which quadruplet each term comes from; `-d N` prints a derivation of any integer N as in Section 4, extending the chain first if N lies beyond the cache.

Videos. Two narrated explanations accompany this note: *Ladhe's Quad Conjecture: A New Pattern in Prime Quadruplets* (28 min, `https://youtu.be/_XSOJ0Yj77Q`), which covers Sections 3–8, and *Build Any Number From Prime Quadruplets* (11 min, `https://youtu.be/JNWvt3hS540`), which walks through the coverage proposition of Section 4 by hand.

App. An interactive companion to this note is being added to the authors' free iOS app *LE-Games*: `https://apps.apple.com/app/id6767880385`.

Conflict of interest. The authors declare no competing interests related to this note.

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A The first 25 quadruplets

For each member: the quad index k, the member t, the base m_k, the gap D, the additive equation stored by the utility (base first), its length including the base, the minimal pure length K of the gap ("–" for the exceptional gaps), and, when the gap was met before, the quad where its equation was first found. $\mathcal{Q}_1$ is written from royal members only.

k	t	m_k	D	additive equation	len	K	from
1	11			(2*3) + 5	2		
1	13			(2*5) + 3	2		
1	17			7 + 5 + 3 + 2	4		
1	19			(2*7) + 5	2		
2	101	19	82	19 + 17 + 13 + 11 + (5*7) + (2*3)	6	–	
2	103	19	84	19 + 11 + (2*5*7) + 3	4	–	
2	107	19	88	19 + 17 + 13 + 11 + (2*3*7) + 5	6	–	
2	109	19	90	19 + 17 + (2*5*7) + 3	4	–	
3	191	109	82	109 + 17 + 13 + 11 + (5*7) + (2*3)	6	–	2
3	193	109	84	109 + 11 + (2*5*7) + 3	4	–	2
3	197	109	88	109 + 17 + 13 + 11 + (2*3*7) + 5	6	–	2
3	199	109	90	109 + 17 + (2*5*7) + 3	4	–	2
4	821	199	622	199 + 197 + 193 + 191 + 19 + 17 + 5	7	6	
4	823	199	624	199 + 197 + 193 + 191 + 19 + 17 + 7	7	6	
4	827	199	628	199 + 197 + 193 + 191 + 19 + 17 + 11	7	6	
4	829	199	630	199 + 197 + 193 + 191 + 19 + 17 + 13	7	6	
5	1481	829	652	829 + 199 + 197 + 193 + 19 + 17 + 13 + 11 + 3	9	8	
5	1483	829	654	829 + 199 + 197 + 193 + 19 + 17 + 13 + 11 + 5	9	8	
5	1487	829	658	829 + 199 + 197 + 109 + 107 + 19 + 17 + 7 + 3	9	8	
5	1489	829	660	829 + 199 + 197 + 109 + 107 + 19 + 17 + 7 + 5	9	8	
6	1871	1489	382	1489 + 109 + 107 + 103 + 19 + 17 + 13 + 11 + 3	9	–	
6	1873	1489	384	1489 + 193 + 191	3	2	
6	1877	1489	388	1489 + 197 + 191	3	2	
6	1879	1489	390	1489 + 199 + 191	3	2	
7	2081	1879	202	1879 + 199 + 3	3	2	
7	2083	1879	204	1879 + 199 + 5	3	2	
7	2087	1879	208	1879 + 197 + 11	3	2	
7	2089	1879	210	1879 + 199 + 11	3	2	
8	3251	2089	1162	2089 + 829 + 199 + 109 + 17 + 5 + 3	7	6	
8	3253	2089	1164	2089 + 829 + 199 + 109 + 19 + 5 + 3	7	6	
8	3257	2089	1168	2089 + 829 + 199 + 109 + 19 + 7 + 5	7	6	
8	3259	2089	1170	2089 + 829 + 199 + 109 + 19 + 11 + 3	7	6	
9	3461	3259	202	3259 + 199 + 3	3	2	7
9	3463	3259	204	3259 + 199 + 5	3	2	7
9	3467	3259	208	3259 + 197 + 11	3	2	7
9	3469	3259	210	3259 + 199 + 11	3	2	7
10	5651	3469	2182	3469 + 2081 + 101	3	2	
10	5653	3469	2184	3469 + 2083 + 101	3	2	
10	5657	3469	2188	3469 + 2087 + 101	3	2	
10	5659	3469	2190	3469 + 2089 + 101	3	2	
11	9431	5659	3772	5659 + 3469 + 199 + 101 + 3	5	4	
11	9433	5659	3774	5659 + 3469 + 199 + 103 + 3	5	4	
11	9437	5659	3778	5659 + 3469 + 199 + 107 + 3	5	4	
11	9439	5659	3780	5659 + 3469 + 199 + 109 + 3	5	4	
12	13001	9439	3562	9439 + 3461 + 101	3	2	
12	13003	9439	3564	9439 + 3463 + 101	3	2	
12	13007	9439	3568	9439 + 3467 + 101	3	2	
12	13009	9439	3570	9439 + 3469 + 101	3	2	
13	15641	13009	2632	13009 + 1489 + 829 + 199 + 107 + 5 + 3	7	6	
13	15643	13009	2634	13009 + 1489 + 829 + 199 + 109 + 5 + 3	7	6	
13	15647	13009	2638	13009 + 1489 + 829 + 199 + 109 + 7 + 5	7	6	
13	15649	13009	2640	13009 + 1489 + 829 + 199 + 109 + 11 + 3	7	6	
14	15731	15649	82	15649 + 17 + 13 + 11 + (5*7) + (2*3)	6	–	2
14	15733	15649	84	15649 + 11 + (2*5*7) + 3	4	–	2
14	15737	15649	88	15649 + 17 + 13 + 11 + (2*3*7) + 5	6	–	2
14	15739	15649	90	15649 + 17 + (2*5*7) + 3	4	–	2
15	16061	15739	322	15739 + 199 + 109 + 11 + 3	5	4	
15	16063	15739	324	15739 + 199 + 109 + 13 + 3	5	4	
15	16067	15739	328	15739 + 199 + 109 + 17 + 3	5	4	
15	16069	15739	330	15739 + 199 + 109 + 19 + 3	5	4	
16	18041	16069	1972	16069 + 1871 + 101	3	2	

k	t	m_k	D	additive equation	len	K	from
16	18043	16069	1974	16069 + 1873 + 101	3	2	
16	18047	16069	1978	16069 + 1877 + 101	3	2	
16	18049	16069	1980	16069 + 1879 + 101	3	2	
17	18911	18049	862	18049 + 829 + 19 + 11 + 3	5	4	
17	18913	18049	864	18049 + 829 + 19 + 13 + 3	5	4	
17	18917	18049	868	18049 + 829 + 19 + 17 + 3	5	4	
17	18919	18049	870	18049 + 829 + 19 + 17 + 5	5	4	
18	19421	18919	502	18919 + 199 + 197 + 103 + 3	5	4	
18	19423	18919	504	18919 + 199 + 197 + 103 + 5	5	4	
18	19427	18919	508	18919 + 199 + 197 + 109 + 3	5	4	
18	19429	18919	510	18919 + 199 + 197 + 109 + 5	5	4	
19	21011	19429	1582	19429 + 1481 + 101	3	2	
19	21013	19429	1584	19429 + 1483 + 101	3	2	
19	21017	19429	1588	19429 + 1487 + 101	3	2	
19	21019	19429	1590	19429 + 1489 + 101	3	2	
20	22271	21019	1252	21019 + 829 + 199 + 197 + 19 + 5 + 3	7	6	
20	22273	21019	1254	21019 + 829 + 199 + 197 + 19 + 7 + 3	7	6	
20	22277	21019	1258	21019 + 829 + 199 + 197 + 19 + 11 + 3	7	6	
20	22279	21019	1260	21019 + 829 + 199 + 197 + 19 + 13 + 3	7	6	
21	25301	22279	3022	22279 + 2089 + 829 + 101 + 3	5	4	
21	25303	22279	3024	22279 + 2089 + 829 + 103 + 3	5	4	
21	25307	22279	3028	22279 + 2089 + 829 + 107 + 3	5	4	
21	25309	22279	3030	22279 + 2089 + 829 + 109 + 3	5	4	
22	31721	25309	6412	25309 + 3257 + 1483 + 1481 + 191	5	4	
22	31723	25309	6414	25309 + 3259 + 1483 + 1481 + 191	5	4	
22	31727	25309	6418	25309 + 3259 + 1487 + 1481 + 191	5	4	
22	31729	25309	6420	25309 + 3259 + 1489 + 1481 + 191	5	4	
23	34841	31729	3112	31729 + 2089 + 829 + 191 + 3	5	4	
23	34843	31729	3114	31729 + 2089 + 829 + 193 + 3	5	4	
23	34847	31729	3118	31729 + 2089 + 829 + 197 + 3	5	4	
23	34849	31729	3120	31729 + 2089 + 829 + 199 + 3	5	4	
24	43781	34849	8932	34849 + 5659 + 3259 + 11 + 3	5	4	
24	43783	34849	8934	34849 + 5659 + 3259 + 13 + 3	5	4	
24	43787	34849	8938	34849 + 5659 + 3259 + 17 + 3	5	4	
24	43789	34849	8940	34849 + 5659 + 3259 + 19 + 3	5	4	
25	51341	43789	7552	43789 + 5659 + 1879 + 11 + 3	5	4	
25	51343	43789	7554	43789 + 5659 + 1879 + 13 + 3	5	4	
25	51347	43789	7558	43789 + 5659 + 1879 + 17 + 3	5	4	
25	51349	43789	7560	43789 + 5659 + 1879 + 19 + 3	5	4	

B The ten exceptional gaps

Gaps that are not a sum of distinct pool members at the moment they are first met, with the additive (A), multiplicative (M) and exponential (E1) representations of the gap stored by the utility.

D	k	t	A	M	E1
22	169	1006331	19 + 3	(11*2)	(3^2) + 13
82	2	101	17 + 13 + 11 + (5*7) + (2*3)	(17*3) + (13*2) + 5	(7^2) + (11*3)
84	2	103	11 + (2*5*7) + 3	(17*3) + (13*2) + 7	(7^2) + 17 + 13 + 5
88	2	107	17 + 13 + 11 + (2*3*7) + 5	(17*5) + 3	(7^2) + (13*3)
90	2	109	17 + (2*5*7) + 3	(17*5) + 3 + 2	(7^2) + 17 + 13 + 11
172	435	3512231	109 + 19 + 17 + 13 + 11 + 3	(19*7) + (17*2) + 5	(13^2) + 3
174	435	3512233	109 + 19 + 17 + 13 + 11 + 5	(19*7) + (13*3) + 2	(13^2) + 5
178	435	3512237	109 + 19 + 17 + 13 + 11 + 7 + 2	(19*7) + (17*2) + 11	(11^2) + (19*3)
180	435	3512239	103 + 19 + 17 + 13 + 11 + 7 + 5 + 3 + 2	(19*7) + (17*2) + 13	(13^2) + 11
382	6	1871	109 + 107 + 103 + 19 + 17 + 13 + 11 + 3	(191*2)	(19^2) + (7*3)

C Notation

$\mathcal{Q}_0$	the royal quad $(2,3,5,7)$
$\mathcal{Q}_k$, $k \geq 1$	the k-th prime quadruplet $(p, p+2, p+6, p+8)$, $p \geq 11$
$q_k^{(d)}$	the member of $\mathcal{Q}_k$ ending in digit $d \in \{1,3,7,9\}$
$m_k = q_{k-1}^{(9)}$	the base of $\mathcal{Q}_k$
$D = t - m_k$	the gap of a member t of $\mathcal{Q}_k$
$\mathcal{P}_k = \mathcal{Q}_1 \cup \cdots \cup \mathcal{Q}_{k-1}$	the pool available to $\mathcal{Q}_k$
$\mathcal{R}$, $\mathcal{R}_{\text{flat}}$, $\mathcal{R}_+$	royal values: all (68), flat (38), additive (11)
$\mathcal{R}^{\pm}$	non-negative royal values with $+, -, \times$ (83)
K	minimal length of a representation of a gap
C_k, T_k	integers with a derivation over $\mathcal{Q}_0..\mathcal{Q}_k$; largest T with $[0,T] \subseteq C_k$
S_k	$S_3 = T_3$, $S_k = S_{k-1} + 4p_k + 16$: the proven lower bound for T_k